

Stool Testing: Infections and Malabsorption.

Consider three children seen in the same week. An eighteen-month-old boy has had five days of watery diarrhoea after a course of amoxicillin for an ear infection. The emergency department sent a multiplex stool panel, and it came back positive for norovirus and for *Clostridioides difficile*. A seven-month-old girl with loose stools had a *C. difficile* test sent by her primary care doctor, and both the antigen and the toxin were positive. And a six-month-old infant with months of watery diarrhoea and poor weight gain has a stool sodium of forty and a potassium of twenty millimoles per litre, with a stool pH of five.

Each of these children has a positive stool result. The question this chapter asks is what each result actually means. The main idea is that a positive stool test is a detection, not a diagnosis. Stool is a sample of everything a child has swallowed, made, shed and carried. Modern tests are so sensitive that they find something in most samples, including samples from healthy children. The skill is knowing whether what was found explains the illness, and knowing when not to send the test at all.

We will work through four questions a stool sample can answer. Which pathogen, if any, is causing the diarrhoea? Is *C. difficile* the cause, especially in a child under two? Is the diarrhoea osmotic or secretory? And is the gut losing protein or fat? For each, we will look at the mechanism, the evidence, the guidelines and the pitfalls, and close with what to do differently in clinic.

A note on the sources. For infectious diarrhoea, the main document is the 2017 guideline of the Infectious Diseases Society of America, called IDSA, written by Shane and colleagues, which covers both children and adults. For *C. difficile*, the paediatric recommendations come from the 2017 guideline of IDSA and the Society for Healthcare Epidemiology of America, called SHEA, published in 2018 with McDonald as first author, and from a 2013 policy statement of the American Academy of Pediatrics, called AAP. For acute gastroenteritis in European children we use the 2014 guideline of the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, called ESPGHAN, and the European Society for Paediatric Infectious Diseases, written by Guarino and colleagues. For adults, we refer to the 2021 *C. difficile* guideline of the American College of Gastroenterology, called ACG, the 2021 focused update from IDSA and SHEA, and the 2021 guidance of the European Society of Clinical Microbiology and Infectious Diseases, called ESCMID. For stool chemistry, the classic reference is the 1999 technical review on chronic diarrhoea from the American Gastroenterological Association, called AGA, and for adults in the United Kingdom, the 2018 chronic diarrhoea guideline of the British Society of Gastroenterology, called BSG.

Part one. Which pathogen? Multiplex PCR and the problem of carriage.

A multiplex polymerase chain reaction panel, often shortened to a multiplex PCR panel, tests one stool sample for the nucleic acid of about twenty viruses, bacteria and parasites at once. This is a great advance in speed and breadth. But the test detects genetic material, not living organisms causing disease. It finds organisms that a child is carrying without symptoms, organisms that are

dead, and organisms still being shed weeks after an illness has ended. It often finds more than one at a time.

The best way to understand the problem is to look at how often healthy children carry the same organisms. In a United States study published in the *New England Journal of Medicine* in 2013, norovirus was found in twenty-one percent of children under five with acute gastroenteritis, but also in four percent of healthy children of the same age. In a large American study of adenovirus, it was found in about nineteen percent of children with gastroenteritis and about seven percent of healthy controls. In a case-control study from the United Arab Emirates, at least one pathogen was found in about forty-three percent of children with diarrhoea, but also in about twenty-three percent of children without it. Enteropathogenic *Escherichia coli*, called EPEC, was found in about eighteen percent of cases and eight percent of controls. In that study, none of the thirteen bacterial targets on the panel, including *C. difficile*, *Campylobacter* and EPEC, was significantly more common in sick children than in well ones, and more than one organism was found in about twenty-eight percent of children.

If carriage is common, how do we decide what is causing the illness? The answer from the largest study of childhood diarrhoea is the amount of organism. The Global Enteric Multicenter Study, called GEMS, enrolled more than nine thousand children under five with moderate to severe diarrhoea and more than thirteen thousand matched community controls in Africa and Asia. In 2016, Liu, Platts-Mills and colleagues reanalysed over five thousand matched pairs with quantitative PCR, which measures how much of each organism is present. For each pathogen they found the quantity above which it was truly associated with diarrhoea. With this approach, a pathogen could be blamed for about eighty-nine percent of cases, compared with about fifty-two percent using the original methods. Six pathogens explained about seventy-eight percent of the attributable diarrhoea: *Shigella*, rotavirus, adenovirus types forty and forty-one, heat-stable toxin producing *E. coli*, *Cryptosporidium* and *Campylobacter*.

The thresholds varied enormously between organisms. For cholera, a few thousand copies per gram of stool were enough. For typical EPEC, it took nearly three billion copies per gram before the organism could be counted as the cause. In other words, EPEC is so common in children that only a very heavy load means anything. Two or more attributable pathogens were present in about thirty-nine percent of cases, which tells us that co-infection is real but must also be interpreted with care.

The type of organism matters as well as the amount. In the American adenovirus study, the enteric types forty and forty-one were almost always the cause when found, with an attributable fraction of ninety-six percent. The non-enteric types, which also cause respiratory infections, were the cause only about half the time, with an attributable fraction of fifty-two percent. A qualitative panel that simply reports adenovirus, without the type, leaves this question open.

What happens when hospitals adopt these panels? A study by Cotter and colleagues in *Pediatrics* in 2021 compared more than eight thousand encounters before and after a children's hospital switched to a multiplex panel. The proportion of positive stool tests rose from eleven percent to forty

percent. The number of stool tests rose by about a fifth. The proportion of children treated for a bacterial or parasitic pathogen roughly doubled, from about two percent to about four percent. But the overall length of stay and the overall hospital charges did not change. Only in the small group of children who received targeted treatment did the stay become shorter, from about five days to about three. A Utah study of more than a thousand paediatric episodes found that just over half of panels were positive, that toxigenic *C. difficile* was the single most common detection at sixteen percent, and that *C. difficile* was found in about one in five infants under a year old. That study had no healthy controls, so we cannot tell how many of those detections were carriage. A panel buys speed and breadth. It does not buy interpretation.

Part two. When should the stool go to the laboratory?

The IDSA guideline of 2017 gives clear triggers. Stool should be tested for enteric pathogens when diarrhoea comes with fever, bloody or mucoid stools, severe abdominal cramping or tenderness, or signs of sepsis. This is a strong recommendation. In contrast, the guideline notes that laboratory testing is generally not needed in an otherwise healthy person with watery diarrhoea, and the ESPGHAN guideline of 2014 says the same for children with acute gastroenteritis: investigations are generally not needed.

The IDSA guideline adds several practical rules. Results of a multiplex panel should always be interpreted together with the clinical picture. When a bacterial pathogen is detected by a culture-independent test, the laboratory should also try to culture it when an isolate is needed for public health, for example to type Shiga toxin producing *E. coli* or *Salmonella*. A traveller whose diarrhoea lasts fourteen days or more should be tested for parasites. And testing again after the symptoms have resolved, a so-called test of cure, is not recommended, except when public health rules require clearance, for example for a food handler.

On these points the paediatric and adult guidance are aligned. Both are restrictive, and the European paediatric guidance is the most restrictive of all.

Part three. *C. difficile* in young children.

C. difficile is a spore-forming anaerobe. Disease is caused by two toxins, A and B, that damage the colonic lining. But many infants carry the organism, including toxin-producing strains, without any illness at all. A meta-analysis by Tougas and colleagues in *JAMA Pediatrics* in 2021 pooled ninety-five studies and more than nineteen thousand children. In healthy infants aged six to twelve months, about forty-one percent carried *C. difficile*, and about fourteen percent carried a toxigenic strain. In children aged five to eighteen years, carriage had fallen to about twelve percent. Why infants carry toxigenic strains without colitis is not fully understood, but the leading explanation is that the immature infant gut lacks the receptors the toxins need to bind and enter cells.

This is why the guidelines are so cautious about testing young children. The IDSA and SHEA guideline says that testing in infants aged twelve months or younger should never be routine, as a strong recommendation. Between one and two years, testing should not be routine either, and should be considered only after other causes have been excluded. From two years, testing is

reasonable when diarrhoea is prolonged or worsening and there are risk factors, such as recent antibiotics or healthcare exposure. The AAP statement of 2013 is a little stricter. It allows testing under one year only in children with Hirschsprung disease, a severe motility disorder, or during an outbreak, and it says that between one and three years a positive result means only possible *C. difficile* infection, so other causes must still be sought. Above three years, a positive result means probable infection. The IDSA infectious diarrhoea guideline places the line at two years. These are differences in detail, but they point the same way: the younger the child, the less a positive test means.

Which test should be used? Nucleic acid amplification tests, called NAAT, detect the toxin gene and are very sensitive, but a positive NAAT cannot distinguish colonisation from disease. The glutamate dehydrogenase antigen test, called GDH, detects the organism, and toxin enzyme immunoassays detect free toxin in the stool, which is closer to active disease. The most useful evidence comes from adults. In 2015 Polage and colleagues studied more than fourteen hundred hospitalised adults. Those who were PCR positive but toxin negative had no *C. difficile* related complications at all, compared with about seven point six percent of those who were positive for both. *C. difficile* related deaths were about point six percent in the toxin negative group and about eight point four percent in the toxin positive group. And diarrhoea lasted a median of two days compared with three. The IDSA and SHEA guideline therefore recommends that, unless a laboratory has strict criteria for which stools it accepts, testing should use a multistep algorithm that includes a toxin test, rather than NAAT alone. The ACG guideline of 2021 says much the same: NAAT or GDH first, then a toxin test. If this is true in adults, it applies even more to a colonised toddler, where a lone positive PCR deserves even less weight. All the guidelines agree that the test should not be repeated within seven days in the same episode, and that there is no role for a test of cure.

Now treatment. For children, the IDSA and SHEA guideline of 2017 recommends either metronidazole or oral vancomycin for a non-severe first episode or first recurrence, and oral vancomycin for severe disease or for a second recurrence. The AAP statement of 2013 still lists metronidazole first for mild to moderate disease. In adults, practice has moved on. The 2021 IDSA and SHEA update and the 2021 ESCMID guideline prefer fidaxomicin, a narrow-spectrum macrolide that spares much of the normal flora, and the ACG accepts vancomycin or fidaxomicin, keeping metronidazole only for low-risk patients.

The paediatric evidence for fidaxomicin comes from the SUNSHINE trial, published by Wolf and colleagues in 2020. One hundred and forty-two children under eighteen were treated, thirty of them under two years of age, after randomisation in a ratio of two to one to fidaxomicin or vancomycin. The initial clinical response was similar, about seventy-eight percent with fidaxomicin and about seventy-one percent with vancomycin. But global cure at thirty days, meaning response without recurrence, was about sixty-eight percent with fidaxomicin and fifty percent with vancomycin, an adjusted difference of about nineteen percentage points. The advantage came from fewer relapses. Fidaxomicin is approved in the United States for children from six months of age. This is a material difference between paediatric and adult guidance. The paediatric evidence has moved past

the paediatric guideline, and children can still receive metronidazole in situations where adults would not.

For repeated recurrences, faecal microbiota transplantation, called FMT, is an option. The largest paediatric cohort, reported by Nicholson and colleagues, included three hundred and seventy-two children at eighteen centres. A single transplant was successful in eighty-one percent, and one or more transplants in about eighty-seven percent. Serious adverse events occurred in about five percent. Fresh rather than frozen stool, and delivery by colonoscopy, were associated with success. The joint NASPGHAN and ESPGHAN position paper of 2019 points out that no controlled trial of FMT has been done in children. Before a first transplant, it is worth confirming a toxin-positive recurrence and considering whether the child actually has inflammatory bowel disease.

Part four. Osmotic or secretory? The stool osmotic gap.

When a child has chronic watery diarrhoea, the stool water itself can tell us why. Stool water leaves the colon at roughly the same osmolality as plasma, about two hundred and ninety milliosmoles per kilogram. That osmolality is made up of electrolytes, mainly sodium and potassium with their accompanying anions, and of anything else that is dissolved, such as unabsorbed sugars, magnesium, sulphate or phosphate.

In secretory diarrhoea, the gut is actively secreting salt and water, so almost all the osmolality is accounted for by sodium and potassium. In osmotic diarrhoea, something that is not absorbed, such as lactose in a child with lactase deficiency, holds water in the lumen, and the electrolytes account for only part of the osmolality.

The osmotic gap is calculated as two hundred and ninety minus two times the sum of stool sodium and potassium. We multiply by two to account for the anions that accompany each cation. According to the AGA technical review, a gap above one hundred and twenty-five suggests a purely osmotic diarrhoea, a gap below fifty suggests a secretory diarrhoea, and values in between suggest a mixed picture. Other sources use slightly different cut-offs. A widely cited review of chronic diarrhoea in infants by Thiagarajah and colleagues in 2018 uses a gap above one hundred for what it calls diet-induced diarrhoea, and the Mayo Clinic laboratory treats a gap above fifty as evidence of an osmotic component. The calculation is the same everywhere.

Why use the fixed number two hundred and ninety rather than the measured stool osmolality? Because the measured value rises in the collection pot within minutes, as bacteria ferment carbohydrate into smaller molecules. A high measured osmolality is therefore unreliable. A low measured osmolality, on the other hand, is useful. A value below two hundred and ninety, or below about two hundred and twenty by the Mayo criteria, means that water or dilute urine has been added to the stool, which raises the question of contamination or of factitious diarrhoea.

Take the two examples from the slide. A child with secretory diarrhoea has a stool sodium of one hundred and ten and potassium of thirty. Two times one hundred and forty is two hundred and eighty, so the gap is only ten. A child with osmotic diarrhoea has a sodium of thirty and potassium

of twenty-five. Two times fifty-five is one hundred and ten, so the gap is one hundred and eighty. Two electrolytes and one subtraction sort the diarrhoea into salt or sugar.

Other simple stool tests support the gap. A stool pH below about five point three suggests carbohydrate malabsorption, because bacteria ferment the unabsorbed sugar into acids, while a pH above about five point six argues against carbohydrate being the only problem. Reducing substances above about half a percent suggest unabsorbed glucose, galactose or lactose. Sucrose is not a reducing sugar, so it is missed unless the sample is first hydrolysed. A stool chloride above ninety millimoles per litre suggests congenital chloride diarrhoea, which is caused by variants in the gene SLC26A3. A fasting trial is also informative. Diarrhoea that stops when the child is fasted points to an osmotic, diet-induced cause, while diarrhoea that persists despite fasting points to a secretory or transport defect, which in an infant should prompt electrolyte studies, endoscopy and genetic testing. And a gap above about seventy-five can point to magnesium, for example from a laxative.

Part five. Losing protein or fat.

Alpha-1 antitrypsin, called A1AT, is a plasma protein that leaks into the gut together with albumin when the gut is losing protein. Unlike albumin, it resists digestion by intestinal enzymes, so it reaches the stool intact. That makes its stool clearance a good marker of protein-losing enteropathy. In the original validation study by Florent and colleagues in 1981, the clearance of alpha-1 antitrypsin correlated very closely with the reference test, the clearance of chromium-labelled albumin, with a correlation coefficient of zero point nine six. Its sensitivity was about ninety-three percent and its specificity ninety percent.

The clearance is calculated from a timed stool collection and a serum level. Laboratory reference ranges treat a clearance of up to twenty-seven millilitres per twenty-four hours as normal, and protein-losing enteropathy usually gives values above fifty. A random stool concentration is only a surrogate. Two pitfalls matter. First, alpha-1 antitrypsin is destroyed by acid below a pH of about three, and in the original study none could be detected in acidic gastric juice. So protein loss from the stomach, as in Ménétrier disease, can be missed unless the test is repeated while the child is on acid suppression. Second, blood in the stool and contamination with urine can raise the result. For the child with low albumin and oedema after a Fontan operation, the right test is a timed clearance, not a random pot.

For fat malabsorption, the classic test is a seventy-two hour faecal fat collection. According to the AGA review, more than seven grams of fat a day is abnormal, but values between seven and fourteen grams are not specific, because any diarrhoea can cause mild steatorrhoea. Values of fourteen grams or more are more specific, and a fat concentration of nine and a half grams or more per hundred grams of stool suggests a pancreatic or biliary cause. The acid steatocrit is a simpler spot test, but in a Polish study the ninety-fifth centile fell from about twenty-four percent at one month of age to about thirteen percent at two years, and the authors warned that it does not reflect faecal fat well in infants. In the United Kingdom, the BSG guideline of 2018 notes that three-day faecal fat collections are no longer performed, and recommends faecal elastase when fat

malabsorption is suspected. As we discussed in the previous lesson, elastase can be falsely low in liquid stool, because water dilutes it.

Finally, parasites. Microscopy for ova, cysts and parasites needs three fresh stools and has a sensitivity of only about sixty to ninety percent. A Giardia stool antigen test has a sensitivity of about ninety-two percent and a specificity of about ninety-eight percent, and antigen tests and PCR panels have largely replaced microscopy. Cryptosporidium is not part of routine microscopy and must be requested specifically, or detected on a PCR panel.

Part six. Where the guidelines agree and where they do not.

Putting the documents side by side, there is agreement on when to send a stool panel or culture, on using a toxin-based multistep test for *C. difficile*, and on avoiding repeat testing and tests of cure. There is a difference in detail on the age below which *C. difficile* results should be distrusted, with IDSA and SHEA drawing the line at one and two years and the AAP extending caution to three years. There is a difference in the cut-off for the osmotic gap, although every source uses the same formula. There is a material difference on first-line treatment, where children may still be given metronidazole while adult guidelines prefer fidaxomicin. And for faecal alpha-1 antitrypsin there is no guideline at all, only laboratory practice.

Part seven. What to do differently in clinic.

Let us return to the three children. The eighteen-month-old with norovirus and *C. difficile* on a panel after amoxicillin almost certainly has norovirus gastroenteritis. At his age, *C. difficile* testing is only valid once other causes are excluded, and here another cause has been found. He needs fluids and time, not *C. difficile* treatment. The seven-month-old with a positive antigen and toxin test should not have been tested. In a child of that age, the result describes her microbiome, not her diarrhoea. She should not be treated for *C. difficile*, and the search should turn to viruses, diet and other causes. The six-month-old with a stool sodium of forty, potassium of twenty and pH of five has an osmotic gap of two hundred and ninety minus one hundred and twenty, which is one hundred and seventy. That is an osmotic, carbohydrate-driven picture. The next steps are to test for reducing substances and to remove the suspected sugar from the diet.

Two more scenarios complete the picture. A nine-year-old with Crohn's disease and a toxin-positive, non-severe *C. difficile* infection has true infection in a high-risk child, and should receive oral vancomycin or fidaxomicin rather than metronidazole. A four-year-old with bloody diarrhoea and fever needs a panel or culture, with culture for Shiga toxin producing *E. coli* if it is detected, and should not receive antibiotics until that organism has been excluded.

The habits to take away are these. Send stool tests only when the result will change management: fever, blood, severe illness, sepsis, an outbreak or prolonged travel-related diarrhoea. Read every positive panel against the carriage rate in healthy children, and remember that load and type matter. Do not test for *C. difficile* under one year, and test between one and two only when other causes have been excluded. Trust the toxin, not the gene. Consider fidaxomicin for children with *C. difficile*, especially when recurrence is a concern. For chronic watery diarrhoea, send stool sodium,

potassium and pH, and calculate the gap from two hundred and ninety. And for suspected protein loss, send a timed alpha-1 antitrypsin clearance, repeated on acid suppression if gastric loss is possible.

Questions to test yourself.

Question one. In the US study of children under five, how often was norovirus found in healthy controls, and what does that mean for a positive panel? Answer: in about four percent. A positive result in a child with diarrhoea is likely but not certain to be causal, and the clinical picture must decide.

Question two. In the quantitative reanalysis of GEMS, how much typical EPEC was needed before it counted as the cause of diarrhoea? Answer: about two point seven billion copies per gram of stool, the highest threshold of any pathogen.

Question three. What proportion of healthy infants aged six to twelve months carry *C. difficile*, and what proportion carry a toxigenic strain? Answer: about forty-one percent and about fourteen percent.

Question four. According to the IDSA and SHEA guideline, when may *C. difficile* be tested in a child between one and two years old? Answer: only after other causes of diarrhoea have been excluded, and never routinely.

Question five. What happened to adults who were PCR positive but toxin negative in the Polage study? Answer: they had no *C. difficile* related complications, compared with about seven point six percent of those who were toxin positive.

Question six. What was the main result of the SUNSHINE trial? Answer: global cure at thirty days was about sixty-eight percent with fidaxomicin and fifty percent with vancomycin, because fewer children relapsed.

Question seven. A stool has a sodium of one hundred and ten and a potassium of thirty. What is the osmotic gap and what does it suggest? Answer: two hundred and ninety minus two hundred and eighty, which is ten. It suggests a secretory diarrhoea.

Question eight. Why can faecal alpha-1 antitrypsin miss protein loss from the stomach, and what can be done about it? Answer: it is destroyed by acid below a pH of about three. Repeat the test while the child is on acid suppression.

End of chapter.